

Persistence Mechanisms

Cyber Security – Complete Technical Guide

Yeh manual 'Persistence Mechanisms' ko samajhne aur implement karne ke liye aik mukammal guide hai. Ise asaan Roman Urdu mein tarteeb diya gaya hai taake aap har technical step ko behtar tor par samajh sakein. Chahey aap cybersecurity ke student hon ya apni lab mein practice kar rahe hon, yeh manual aapke study notes aur projects ke liye aik behtareen reference sabit hoga.

CYBER SECURITY

BEGINNER FRIENDLY

ROMAN URDU

Persistence Mechanism Kya Hai?

Basic Definition

Jab koi attacker kisi computer ya system mein ghus jaata hai, to uska pehla maqsad hota hai ke **wapis aana aasaan ho** — chahe system restart ho jaaye yhn wapis any se means he ke restart ke bad bi wo usi ke computer mein ho , reboot ho jaaye, ya user log out kar le.

Is maqsad ke liye jo techniques istemal hoti hain, unhein **Persistence Mechanisms** kehte hain.

Real Life Example

Sochiye kisi chor ne aapke ghar ki **chabi ki copy** bana li. Ab chahe aap darwaza band karein ya khidkiyan band karein — chor kabhi bhi wapis aa sakta hai.

Computer mein bhi waisa hi hota hai. Attacker apni "chabi" system mein chupa leta hai taake woh kabhi bhi wapis aa sake.



Persistence = System mein **permanent access** banana

Master Manual: Remote Persistence & Execution

Maqsad: Ek baar target system ka control milne par, use permanent (hamesha ke liye) control mein rakhna, taake system restart hone par bhi access khatam na ho.

Stage 1: Payload Execution (Target ke system par file chalana)

Pehle yeh samjhein ke target ke system par file **kaise** chalti hai.

- Transfer:** Aapne my_file.exe ko target system ke C:\Windows\Temp\ folder mein bheja hai (ya upload kiya hai).
- Execution (File Chalana):** Remote shell mein yeh command likhein:
 - Command:** start C:\Windows\Temp\my_file.exe
 - Result:** Yeh command target system par file ko background mein execute kar degi.
 - Tip:** Agar aap chahte hain ke user ko bilkul pata na chale, to file ko hamesha **hidden** attribute ke sath rakhein. (Command: attrib +h C:\Windows\Temp\my_file.exe)

Stage 2: The Persistence Setup (Hamesha ke liye access)

Jab file chal jaye aur aapka connection (Shell) stable ho jaye, tab yeh steps follow karein:

Step 1: System ko permanent record mein daalna

Target ke system mein Command Prompt (Admin) mein yeh line likhein aur **Enter** dabayein:

```
schtasks /create /tn "Microsoft_System_Update" /tr "C:\Windows\Temp\my_file.exe" /sc onlogon /ru System /f
```

- Samajh:** Humne Windows ko ek aisa "Task" de diya hai jo har baar computer on hone par (onlogon) hamari file ko System ki power (highest privilege) ke sath khud-ba-khud chala dega.

Step 2: Task confirm karna

Yeh check karein ke task list mein sahi tarah se add hua ya nahi:

```
schtasks /query /tn "Microsoft_System_Update"
```

- Samajh:** Agar status "Ready" dikh raha hai, to aapka system bilkul taiyar hai.

Stage 3: Remote Testing (Testing ka Amal)

Ab check karein ke kya sab kuch theek kaam kar raha hai:

- Run Task:** System restart kiye bina check karein:

```
schtasks /run /tn "Microsoft_System_Update"
```
- Check Connection:** Apne main laptop (Attacker Machine) par jayein aur dekhein ke kya **Netcat** ya Metasploit par naya session/connection aa gaya hai.

Master Summary (Step-by-Step Flow)

Step	Action	Command/Method
1	Upload	File ko target ke Temp folder mein bhejen.
2	Execute	start C:\Windows\Temp\my_file.exe
3	Persistence	schtasks /create /tn "Microsoft_System_Update" ...
4	Verify	schtasks /query /tn "Microsoft_System_Update"
5	Cleanup	del C:\Windows\Temp\my_file.exe (Agar zaroorat na ho).

Zaroori Hidayat (Security Professional ke liye):

- Masking (Naam badalna):** Hacker kabhi file ka naam my_file.exe nahi rakhte. Wo hamesha svchost.exe, winlogon.exe ya chrome_updater.exe jaisa naam rakhte hain taake Task Manager mein koi shak na kare.
- Ethical Obligation:** Yeh manual sirf aur sirf **Security Testing** aur **Laboratory Environments** ke liye hai. Kisi bhi aise system par jahan aapke paas ijazat (Written Consent) nahi hai, wahan in commands ka istemal karna ghair-qanooni hai.
- Defence (Bachne ka tareeqa):** Ek pro-level security expert wahi hota hai jo ise detect kar sake. Hamesha Autoruns tool use karein aur kisi bhi aisi service ya task ko delete kar dein jo unknown source ya Temp folder se chal rahi ho.

Cyber Security Mein Importance

Persistence mechanisms ko samajhna har security professional ke liye **bohat zaroori** hai. Yeh jaanna ke attacker kaise kaam karta hai, aapko behtar defender banata hai.



Attacker Ke Liye

- Long-term access hasil karna
- Data chori karna bina pakde jaaye
- System ko control mein rakhna
- Doosre attacks ke liye base banana



Defender Ke Liye

- Threat detect karna jaldi
- Incident response behtar banana
- System ko secure rakhna
- Forensics mein madad karna



MITRE ATT&CK

- TA0003 — Persistence
- 30+ techniques documented
- Industry standard framework
- Threat intelligence ka hissa

Method 1: Registry Run Keys

T1547.001

Windows Registry mein kuch khaas keys hoti hain jo system start hote hi automatically programs run karti hain. Attackers inmein apna malicious software daal dete hain.

Kaunsi Keys Hoti Hain?

- HKCU\Software\Microsoft\Windows\CurrentVersion\Run
- HKLM\Software\Microsoft\Windows\CurrentVersion\Run
- HKCU...\CurrentVersion\RunOnce

Simple Example

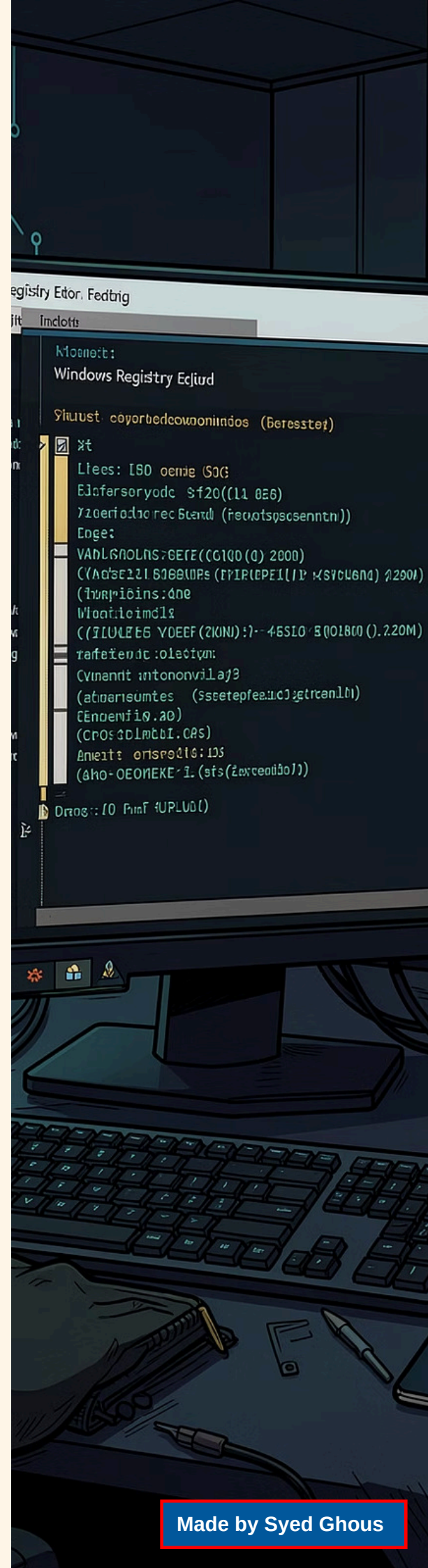
Attacker ye command run karta hai:

```
reg add  
"HKCU\Software\Microsoft\Windows\CurrentVersion\Run" /v "UpdateService" /t  
REG_SZ /d "C:\malware.exe"
```

Ab har baar user login karega, **malware.exe** automatically run hoga — bina user ko pata chale.

Kyun Effective Hai?

- Simple aur aasaan technique
- Koi extra software nahi chahiye
- Reboot ke baad bhi kaam karta hai
- Normal user ko pata nahi chalta



Method 2: Scheduled Tasks

T1053.005

Kya Hai Scheduled Tasks?

Windows mein **Task Scheduler** hota hai jo kisi bhi program ko ek time par ya specific event par automatically run kar sakta hai. Attackers is feature ko misuse karte hain.

Attacker Kaise Use Karta Hai?

- Har 30 minute baad malware run karna
- User login ke waqt script chalana
- System start hone par command run karna
- Random time par data exfiltrate karna

Example Command

```
schtasks /create /tn "WindowsUpdate"  
/tr "C:\malware.exe" /sc onlogon
```

Ye command ek aisi task banata hai jo **"WindowsUpdate"** ke naam se chupegi aur har login par malware run karegi.

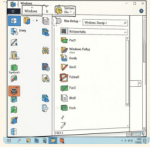
Detection Tip

Task Scheduler open karein aur dekhain ke koi ajeeb ya unknown task to nahi hai — khaas kar agar uska publisher unknown ho.

Method 3: Startup Folder

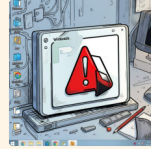
T1547.001

Windows mein ek khaas folder hota hai jisme agar koi shortcut ya file daal di jaaye, to wo har baar system start hone par automatically run hota hai.



Startup Folder Locations

- **User Level:** C:\Users\[Username]\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup
- **All Users:** C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup



Attack Scenario

Attacker apna malicious .exe ya .lnk (shortcut) file is folder mein copy kar deta hai. Ab jab bhi user login karega, malware apne aap chal padega.

Ye technique **bohat purani** hai lekin aaj bhi effective hai kyunke bohat se log is folder ko check hi nahi karte.

Method 4: Windows Services

T1543.003

Windows Services Kya Hain?

Windows Services wo programs hain jo **background mein** chalte rehte hain — bina kisi user interface ke. Ye system start hote hi load ho jaate hain.

Attackers apna malicious code ek **fake service** ki tarah install kar dete hain jo hamesha chalta rahega.

Attacker Kaise Create Karta Hai?

```
sc create "MyService" binPath=  
"C:\malware.exe"  
sc start "MyService"
```

Ab ye service hamesha background mein chalti rahegi aur attacker ko remote access deti rahegi.

Key Points

- Services **SYSTEM level** par chalti hain
- Normal user ko pata nahi chalta
- Antivirus bhi kabhi miss kar deta hai
- Service ka naam legitimate lagta hai (e.g., "WindowsUpdateHelper")

Method 5: WMI (Windows Management Instrumentation)

T1546.003

WMI ek advanced technique hai jo bohat se security tools ko bypass kar sakti hai. Ye **fileless persistence** ka tareeqa hai — matlab koi file disk par nahi likhi jaati.

1

WMI Kya Hai?

Windows Management Instrumentation ek built-in Windows feature hai jo system administration ke liye use hota hai. Attackers isay **abuse** karte hain.

2

Kaise Kaam Karta Hai?

Attacker WMI **Event Consumer** banata hai jo kisi specific event (jaise system start) par malicious code run karta hai — bina koi file create kiye.

3

Kyun Dangerous Hai?

Kyunke ye **memory mein** rehta hai, traditional antivirus ise detect nahi kar paata. Forensics bhi mushkil hoti hai kyunke disk par koi file nahi hoti.

4

Detection Command

PowerShell mein ye command run karein:

```
Get-WmiObject -Namespace  
root\subscription -Class  
__EventConsumer
```

Agar koi suspicious consumer mile to investigation karein.

Detection: Kaise Pata Karein?

Ek security student ke taur par, aapko ye jaanna chahiye ke system mein persistence mechanism detect kaise karte hain. Yeh skills aapko **Blue Team** ya **Incident Responder** banne mein madad karenge.



Registry Check Karein

`regedit` open karein aur Run keys check karein. Koi unknown entry ya ajeeb path dikhe to investigate karein.



Startup Folder Check

`shell:startup` run karein aur dekhein ke koi ajeeb shortcut ya file to nahi hai.



PowerShell History

Attacker aksar PowerShell use karta hai. `Get-History` command se piche commands dekhein.



Scheduled Tasks Dekhein

Task Scheduler mein jaakar saari tasks review karein. Unknown publisher ya suspicious command par ghor karein.



Services Inspect Karein

`services.msc` open karein. Unknown services, especially jo "Automatic" start type ki hon, unhein verify karein.



Network Monitor

Wireshark ya Windows Firewall logs check karein. Koi unknown outbound connection suspicious ho sakta hai.

Mitigation: Bachne Ke Tareeqe

Persistence attacks se bachne ke liye aapko **layered security approach** leni chahiye. Ek hi solution kaafi nahi — multiple defenses ek saath kaam karte hain.

1

Least Privilege

Users ko sirf utni access dein jitni zaroori ho.
Admin rights har kisi ko na dein.

2

Regular Audits

Registry, Services, aur Scheduled Tasks ka
regular review karein. Koi naya entry to nahi?

3

EDR Solution

Endpoint Detection and Response tools install
karein jo suspicious behavior detect karein.

4

Disable Unused Features

WMI aur PowerShell ko restrict karein agar
business ko zaroorat na ho.

Quick Reference Checklist

- ✓ Admin accounts strong passwords use karein
- ✓ Antivirus hamesha updated rakhein
- ✓ Windows updates regularly install karein
- ✓ Unnecessary services disable karein
- ✓ Log monitoring enable rakhein
- ✓ Backup regularly lete rahein

Yaad Rakhein

Persistence mechanism samajhna sirf
attack ke liye nahi — ye aapko **behtar
defender** banata hai.

MITRE ATT&CK framework mein
TA0003 ko regularly study karte
rahein.

✓ Knowledge = Power 🔑